



CHAPTER 10

The Entrepreneurial ‘Mindset’: Entrepreneurial Intentions from the Entrepreneurial Event to Neuroentrepreneurship

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BORN OR MADE?

In the early days of entrepreneurship, we were plagued by research that lacked strong theoretical grounding. Our curiosity took us in remarkable directions as we explored fascinating questions about entrepreneurship and entrepreneurs. However, too often we produced striking results that were ad hoc and not replicable. As the 1980s were ending, use of even the basics of social psychology was scarce. The one academic domain that we did call upon was personality psychology. Unfortunately, personality psychology had long since gone into eclipse, and, moreover, we often had deployed it unskillfully. In particular, we had committed ourselves to an essentially trait-based model of entrepreneurs.

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Why? Because we shared the romantic (and intuitively ‘obvious’) notion that entrepreneurs are somehow magical, we assumed that entrepreneurs are special people with unique properties. On the other hand, psychology had progressed with growing recognition that these seemingly fixed traits are much more malleable and highly situational. That is, entrepreneurship is a function of both person and situation.

In retrospect, applying formal models of behavioral intentions was an excellent leverage point. Social psychology offered what appeared to be a solid conceptual framework along with robust empirical tools and considerable statistical power. Meanwhile, one of our field’s founders, Albert Shapero, had independently formulated his model of the “entrepreneurial event” (Shapero 1982). He argued that a credible opportunity was driven by perceptions of desirability and feasibility. Similarly, the dominant intentions model (the theory of planned behavior, in short: TPB) was driven by perceptions of personal and social desirability and of controllability (feasibility).

Given the similarities of the models, it was less difficult to sell TPB to reviewers. However, the very first attempt (Krueger 1989) took three years to get empirical results published (Krueger 1993) and nearly four years for a conceptual article (Krueger and Carsrud 1993). Both articles continue to be cited along with a follow-up that expanded the model to organizational settings (Krueger and Brazeal 1994).

Despite the immediate, wide embrace of this fairly basic model from social psychology, acceptance of deep theory and rigorous methods from psychology by entrepreneurship scholars took some time. But ultimately, those scholars started adopting tools from social and eventually cognitive psychology. The increase in entrepreneurial intention (EI) research during the 1980s and 1990s coincides with the introduction of contributions from the field of psychology (Fayolle and Liñán 2014). Ever since then, the concept of EIs is recognized to be important for our understanding of entrepreneurial behavior (Shane and Venkataraman 2000; Fitzsimmons and Douglas 2011). EIs affect entrepreneurial behavior and, subsequently, venture development and growth (Bird 1988) and they are among the most researched phenomena in the field of entrepreneurship (Liñán and Fayolle 2015; Gupta et al. 2016).

But as we delved more deeply into human thinking, we slowly realized that we also need to dive more deeply than intentions; we needed to understand “what lies beneath” (Krueger 2007). Insights into the

development of EIs facilitate our understanding of how entrepreneurial cognition and, ultimately, how the entrepreneurial mindset develops (Krueger 2007). Therefore, the more we study intentions, the more we need to look deeper at where intent actually arises.

The EI literature is huge (as of this writing, Google Scholar lists over 1260 articles with that phrase in the publication title and more than 16,000 overall), and it is the rare scholar who is not familiar who has little or no familiarity. However, the trajectory of the topic is often overlooked. From whence do intentions arise? We will use this opportunity to explore some research where intentions research not only opened the door but also reciprocates by contributing to future intentions research.

In this chapter, we first describe the most influential modeling and application of this concept in the entrepreneurship literature. Then, we demonstrate how this concept has affected entrepreneurship studies throughout the last 30 years. In addition, we point to the links with current studies that are relevant for understanding entrepreneurial opportunity identification, development and exploitation. Then we zoom in on a specific promising future research alley, namely the role of entrepreneurship education in shaping EI, which we discuss from a sense-making perspective.

PSYCHOLOGICAL FOUNDATION OF OPPORTUNITY DEVELOPMENT

Intentions as an intellectual topic is millennia old (e.g., Socrates). However, it was Al Shapero (e.g., Shapero & Sokol 1982) who brought us a well-developed conceptual model of how entrepreneurial intent evolves in his model of the entrepreneurial event. He focused on how intent reflects a significant and actionable opportunity. The term ‘entrepreneurial intentions’ emerged in the late 1980s. In turn, the emergence of EI is linked to an increasing focus on psychological and cognitive aspects of entrepreneurship.

Perhaps the seminal article is from Bird (1988) for whom the intentional (entrepreneurial) process begins with a person’s needs, values, wants, habits and beliefs. By pointing out the influence of psychological processes in entrepreneurship, Bird’s study lay the groundwork for a cognitive approach to the study of entrepreneurship. Bird (1988) points out that a person’s intentions and actions are structured by rational, analytic and cause-and-effect-oriented thinking, while intuitive, holistic and

contextual thinking frame intentions and actions. Bird makes a clear distinction between EIs and concepts such as goal setting and the manager's intentions in established firms. By doing so, she helped establish EI as a separate field of research with its own theoretical framework which numerous later studies built upon (Gupta et al. 2016). In those later studies, EIs are seen as a function of psychological variables, such as perceptions of desirability, feasibility or behavioral control; however they differ with regard to conceptual modeling as well as empirical application in the field of entrepreneurship.

According to Bird (1988), intentionality draws on two important variables: alignment and attunement. Alignment and attunement can result in more effective entrepreneurial action and, subsequently, more effective new ventures. Alignment involves the configuration of parts in a way that they are contributing to a single purpose and direction. Alignment thus expresses intentionality because it makes the “many inner voices that reflect different and conflicting needs, values, memories and wishes agree on one direction” (Bird 1988: p. 449). Conflict in values can slow, stop or divert action; therefore for entrepreneurs, aligning individuals on the entrepreneurial team and the early staff can be an important and difficult leadership task (Bird 1988).

Attunement relates to one's readiness to send and receive information, influence or meaning from other sources; it requires vigilance, open-mindedness, extroversion and the ability to learn from mistakes (Bird 1988). Attunement draws on the personal values and beliefs of the entrepreneur and the impact those values and beliefs have on the entrepreneur's organizing capabilities. Forms of attunement are found in the entrepreneur's networking with external team members such as bankers, venture capitalists, lawyers, accountants and consultants in order to gain key resources for the new venture (Bird 1988). As we see later, this becomes particularly important as we move concretely to see “what lies beneath”.

The Theory of Planned Behavior

A first rigorous and empirically testable conceptualization of EIs appeared in 1989 in Krueger's dissertation, which included the full model prescribed in the theory of planned behavior plus variables that captured Shapero's constructs.¹ The first empirical test was published in early 1993, followed by Krueger and Carsrud's (1993) intentions model based on the

theory of planned behavior. As with the dissertation, the authors drew on (Ajzen & Fishbein's 1980) model of behavioral intentions which specifies three distinct attitudinal antecedents of intention, each drawn from existing theory and prior evidence: personal attitude toward outcomes of the behavior, perceived social norms—both together reflect perceived desirability of performing the behavior—and perceived behavioral control (Krueger and Carsrud 1993). We see here how EIs are conceptualized and made measurable not only in terms of cognitive and psychological attributes but also in terms of the entrepreneurs' perceptions of, or consideration for, the social context.

Krueger and Carsrud (1993) opened a new alley for EIs research with an article explaining the potential for theory and research from the theory of planned behavior. Research that drew on Krueger and Carsrud's work or generally used the theory of planned behavior in studies of the formation, effect and modeling of EIs increased after the introduction of Ajzen and Fishbein's psychological approach to entrepreneurship by Krueger & Carsrud in 1993 and the first empirical study (Krueger 1993). Lars Kolvereid and his colleagues have also built an impressive array of entrepreneurship studies using the full-blown TPB model. Liñán has created his own version of the model with a widely used questionnaire, his Entrepreneurial Intentions Questionnaire (EIQ) (Fig. 10.1).

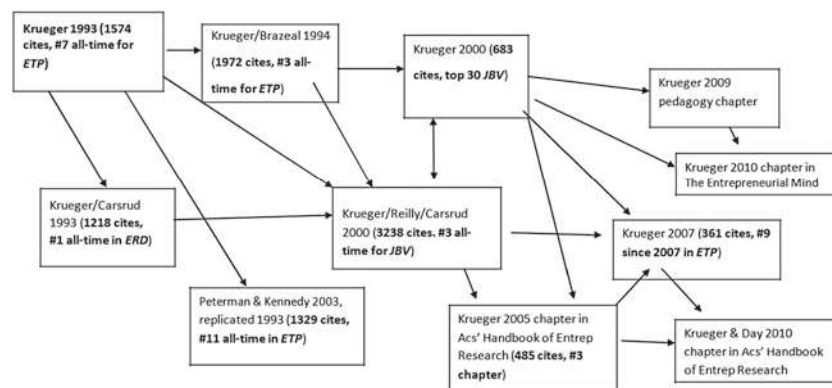


Fig. 10.1 Offers a flow chart of research on intentions by Krueger and colleagues

QUANTIFYING THE IMPACT OF INTENTIONS RESEARCH

The importance of the concept of EIs is reflected in its application in entrepreneurship studies, in particular during the 1990s and after the number of studies on entrepreneurial grew explosively.

Consider these recent statistics from Google Scholar: The number of citations for “entrepreneurial intentions” and “entrepreneurship”.

	<u>Before 1993</u>	<u>After 1993</u>	<u>Growth</u>
Entrepreneurial Intentions	112	~13,500,120X	

(Entrepreneurship research overall only grew 3.7X. That is, intentions research has grown more than 30 times as fast as entrepreneurship research in general.)

We also see a nine-fold increase in entrepreneurial cognition research relative to entrepreneurship in general, although it accelerated considerably after 1993, suggesting EI research was at least a catalyzing factor for research there.

The longevity of Krueger and Carsrud’s (1993) work is striking and continues to represent the most-cited work on this topic. The original 1993 article in *Entrepreneurship Theory & Practice* (ETP) remains the seventh-most cited article in that journal. The 1994 Krueger and Brazeal article is the third-most cited ever in *ETP*. The 2000 article by Krueger, Reilly and Carsrud has over 3000 citations and is the third-most cited article ever in the *Journal of Business Venturing*. And the 1993 Krueger and Carsrud’s work appears to be the #1 most-cited article published in *Entrepreneurship & Regional Development*. All of these have 1000 or more citations.

However, the authors are proudest of the research they have inspired, especially where research has focused on deeper issues. In specific, Krueger has argued that intent may be so easy to conceptualize and measure, but it is only the starting point.

A decade ago, the author and others proposed trying to blow up the intentions model (n.b. to surprising anxiety by colleagues who loved the model). A group of leading intentions scholars pooled their data and tested whether the intentions model was explained better by its purported antecedents or whether it was actually the other way around. As the results showed, once intent became the driving variable, adjusted r-squared went up (Brannback et al. 2007).

Why? Maybe TPB should be a reflective model rather than a formative model. Or, if one asked a philosopher or a cognitive scientist like Bratman (1987), they would point out that the TPB model is a static snapshot of a highly dynamic process (Brannback et al. 2007; Krueger 2017).

Why and How the Ideas Therein Are Still Relevant to the Field

Where is this trajectory of research into EIs leading us? We suggest that if intentionality has an important role in entrepreneurship, as the increase in research suggests, we need to know about the factors that affect it.

In a recent book chapter (2017) Krueger asked if research into EIs was growing or just getting bigger. In other words: Of all those literally thousands of papers that touch on intent, how many are really breaking new ground? The short answer is “Not enough” but those that are suggest a bright future. Here are recent examples of those that do break new ground.

We offer a sampling of intriguing research that have begun to address deeper issues, including some that look at negative influences. For example, Cesinger et al. (2011) look at entrepreneurship's ‘dark side’ (the so-called Dark Triad, narcissism, Machiavellianism and psychopathy). What is particularly interesting here is that this fits Krueger's (2007) call that drawing from developmental psychology will be productive. Adam (2015) examines the role of commitment in the EI-behavior link, by applying the commitment theory of Meyer and Allen (1987, 1991), another area of social psychology that has been woefully neglected. The goal of the study is to assess to what extent, and how, commitment helps them to enact their intentions. The study shows that commitment has a positive impact on entrepreneurial behavior when the approach is motivated by the inner self. However when the project is motivated by external reasons, it seems to have no or even negative impacts on the entrepreneurial process. The results also show that it does not matter if the intended entrepreneur is emotionally attached to entrepreneurship as a career or to his project itself, and neither does being committed to both entrepreneurship and the project increase the probability to create a company.

The aforementioned studies yield insights into the wide array of psychological topics (in entrepreneurship) to which the concept of EI can contribute. Most important, any intent represents the direction and purpose of the action (e.g., Bratman 1987). EIs relate not only to the entrepreneur's decision or disposition to become or be an entrepreneur but also

to day-to-day intentions that the entrepreneur formulates. Entrepreneurial intentionality is linked to everyday decision-making about direction and purpose of the entrepreneur's goal-oriented behavior. Intentionality is linked to mindfulness about assumed—perceived—desirability and feasibility of a directed and purposeful idea for action. All of this suggests the future is bright for not only intentions research but even more so for research that opens new doors.

We continue to see study after study that blindly applies the intentions model to yet another population. However, we are finally starting to see that scholars are explicitly considering the potential impact of, for example, culture. The GUESS research project (Sieger et al. 2011; <http://www.guess-survey.org>) now reaches more than 20 countries, and the Global Entrepreneurship Monitor (GEM) collects data in over 60. (In fact, the GEM team in Spain collected data from 17 different regions within Spain, finding intriguing differences.) And if we delve further and further into 'what lies beneath' intentions, we find ourselves moving in several useful directions.

CHANGING INTENT TO CHANGING THE MINDSET?

Perhaps the most striking direction of EI research is also the most valuable for policymakers and for educators: entrepreneurship education. Is the point of entrepreneurship education to raise the learners' intent toward entrepreneurship? Not necessarily. Do we want to make entrepreneurship more credible long term (more desirable and more feasible) or raise intent in the near term (meaning: do we want to raise the intent of learners who simply are not ready)? If the former, we really do need to move away from intent *per se* and move toward nurturing an entrepreneurial mindset, focusing on changing deep beliefs that ultimately anchor entrepreneurial thinking (Krueger 2007, 2015).

We now hear claims from educators that we are no longer teaching *about* entrepreneurship, we are now teaching them to *do* entrepreneurship (thinking and action). While as of this writing, such education is rarer than we claim and it is hard to assess whether the learning approaches are effective. This calls for four interrelated strategic directions.

First, measuring changes in intent is unlikely to be a productive dependent variable, even though the vast majority of impact studies use exactly that (Nabi et al. 2017). We are in need of much more comprehensive

metrics (Lackeus 2015), including those that address explicitly the entrepreneurial mindset. Where are our rigorous tests of the mindset? Progress is being made but needs to accelerate (Krueger 2015; Penaluna and Penaluna 2015).

Second, as Krueger (2007) pointed out, it is the rare entrepreneurship scholar who has even a limited familiarity with developmental psychology. It is as alien to us as cognitive psychology was before Baron (e.g., 2004). Studying how humans actually learn would also have immense benefits for students. The number of university-level entrepreneurship educators with even the slightest acquaintance with professional educator training is nearly zero. Maybe it is time? And maybe research will be the excuse? Imagine what more we could do with a solid ground in learning theories that we seem to use in classrooms already (Robinson et al. 2016).

Third, diving deep will inevitably take us to truly fundamental cognitive science, even neuroscience (Krueger and Welpé 2014; Day et al. 2017). One nascent example of using cognitive science is the new social ontology model (McBride et al. 2013). This direction is profoundly promising but, again, rarely a part of any entrepreneurship scholar’s training.

Finally, going ‘wide’ is just as important. There is so much in social and cognitive psychology that have not seen widespread adoption in our field, even when it is used productively in other management domains. Let us take a long look at how that is already providing valuable insights. It also provides enormous implications for entrepreneurship education. (And embracing those intentions is a dynamic process.)

Confirmatory Bias and Intentionality of Feedback

The study of EIs is inextricably intertwined with the study entrepreneurial cognition. To answer the question of what is forming EIs, we therefore need to dig deeper by looking at sociocognitive structures. Entrepreneurial cognition relates to interpretative and decision-making processes involved in opportunity identification and development. And EIs are rooted in interpretative (cognitive) processes of what is acceptable practice by relevant others; for example from what Shepherd and Patzelt (2017) call the entrepreneur’s ‘community of inquiry’. In particular, perceptions of social norms—as part of the perceived desirability, which is an antecedent to EIs—require information exchange with third parties who provide information about the legitimacy of an entrepreneurial opportunity.

Through feedback processes, stakeholders provide an entrepreneur with intersubjective knowledge crucial for successful business opportunity development (Venkataraman et al. 2012). Feedback is important for opportunity recognition and development (e.g., Ozgen and Baron 2007), and it is therefore crucial that entrepreneurs are open to new information and knowledge received during that feedback. Intentionality about feedback occurrences helps entrepreneurs to avoid the so-called confirmation bias.

In their everyday cognitive activity, entrepreneurs are prone to various cognitive biases in their information searching and receiving behavior.

Mitchell et al. (2008) show that the confirmation bias negatively affects the openness to new ideas during entrepreneurs' opportunity identification and pursuit: "[P]erceptions are influenced and possibilities are limited when new information is interpreted such that preconceptions are confirmed and learning is suppressed. [...] For instance, where behavior is selected from a narrow range of path dependently acquired routines as opposed to assessing all possible choices (e.g., Miller 2007), the level of achievable expertise may be limited" (Mitchell et al. 2008: p. 228). In other words, the confirmation bias jeopardizes learning because the entrepreneur tends to overlook evidence which opposes his beliefs or assumptions.

The confirmation bias is relevant in the context of entrepreneurial opportunity development because it causes people to reject information that might indicate that their current assumptions are incorrect, whereas they readily accept information with positive connotations (McGrath 1999). In other words entrepreneurs-to-be might shy away from negative or harsh feedback which proves a liability for opportunity development. Because entrepreneurs are not always or not equally willing to accept feedback even from relevant stakeholders, it remains a riveting question how entrepreneurship educators can teach (aspiring) entrepreneurs to be *intentional* about negative feedback—both about being receptive to it and being responsive to it.

This intentional and co-creational approach to feedback engagement by the entrepreneur has been recognized in existing entrepreneurship literature. Fisher (2012), for example, evaluates the behaviors underlying the theories of effectuation, causation and bricolage. He finds that intentional engagement with stakeholders is seen to be crucial in achieving the intended goal—opportunity exploitation—in all three theories (Fisher 2012).

Feedback intentionality is reflected in the entrepreneurs’ efforts to align relevant stakeholders, and attune relevant information with their own values and beliefs about the entrepreneurial opportunity. For entrepreneurs, aligning individuals on the entrepreneurial team and early staff is an important and difficult leadership task, just as is the entrepreneur’s networking with external team members such as bankers and venture capitalists. Similarly, the entrepreneur needs to engage in networking with lawyers, accountants and consultants, in order to gain key resources for the new venture (Bird 1988).

*Developing the Entrepreneurial Mindset: Seeking Feedback
Intentionally ...*

Intentionality about feedback can be observed in entrepreneurial sense-making processes. Sense-making reflects a concern with the acquisition and processing of information in order to establish “a manageable level of uncertainty” (Vlaar et al. 2008). For research on the dynamic properties of EIs, we propose to combine insights from studies on EIs with the sense-making perspective on entrepreneurial cognitive development and conceptualize the entrepreneurial mindset in terms of intentional feedback cycles.

Sense-making consists of interpretative processes about what counts in a situation and why. It consists of bracketing, labeling and categorization activities. It is by conceptual distinction that any perception or impression is made meaningful (Weick et al. 2005), and sense-making enables actors to make pieces of information meaningful by discerning them, ordering them according to their value and usefulness and subsequently arranging them in relation to other pieces of information.

Sense-giving has also been broadly researched in organizational and entrepreneurship studies. It “*consists of acts by which individuals attempt to alter and influence the way others think and act; concerns activities by which stakeholders frame and disseminate visions and beliefs to others so as to increase their understanding and support; may include offering descriptions and explanations, providing signals, constructing credible and consistent narratives, and projecting images through stories, slogans, metaphors, and artifacts*” (Vlaar et al. 2008: p. 240). Sense-giving helps the entrepreneur to convey his opportunity—and propositions—to stakeholders, by means of language to persuade their stakeholders.

While entrepreneurial sense-making and sense-giving have been fruitfully researched in a large number of studies (Hill and Levenhagen 1995; Weick 2005; Clarke and Holt 2010; Cornelissen et al. 2012; Weick 2012), they are interlinked with the less well-researched and less applied concepts of sense-demanding and sense-breaking.

Sense-breaking is a process in which someone's understanding is disrupted by contradictory evidence or values provided by others in the process of sense-making (Vlaar et al. 2008; Pratt 2000). Sense-breaking, according to Vlaar et al. (2008), is a process "*used to question existing understandings of others*" (p. 241) and which involves "*the reframing of previously held conceptions and redirecting [...] attention and search for solutions*" (idem).

Sense-demanding is associated with efforts by individuals to acquire and process information, in order to establish 'a manageable level of uncertainty'. To improve the quality of information upon which they base their decisions, individuals seek as much relevant information as possible (Vlaar et al. 2008). By sense-demanding, the entrepreneur actively and purposefully seeks feedback on his or her opportunity which manifests in information, knowledge or other forms of (im) material support for the preparation and/or exploitation of that opportunity.

We propose that the four interrelated processes of sense-making, sense-giving, sense-demanding and sense-breaking offer a valuable analytical tool in studying the cognitive nature of EIs.

The conceptualization of feedback loops in terms of sense-giving, sense-demanding and sense-breaking occurrences yield a more dynamic analysis of the bidirectional effect of EIs. We know since Bird that intentions are the mediator between attitude and actions, but perhaps intentional feedback loops also affect the antecedents of our intentions, such as deeper beliefs (Krueger 2007). Krueger (2009) already recognized that the dynamic factors of EIs, sense-giving, sense-demanding and in particular sense-breaking, can be used as analytical tools to examine the development of EIs. In this conceptualization, sense-giving reflects alignment activities while sense-demanding helps with the entrepreneur's alignment and attunement activities (see Fig. 10.2).

Below we describe in more detail the way that sense-breaking is shaped by and shapes EIs.

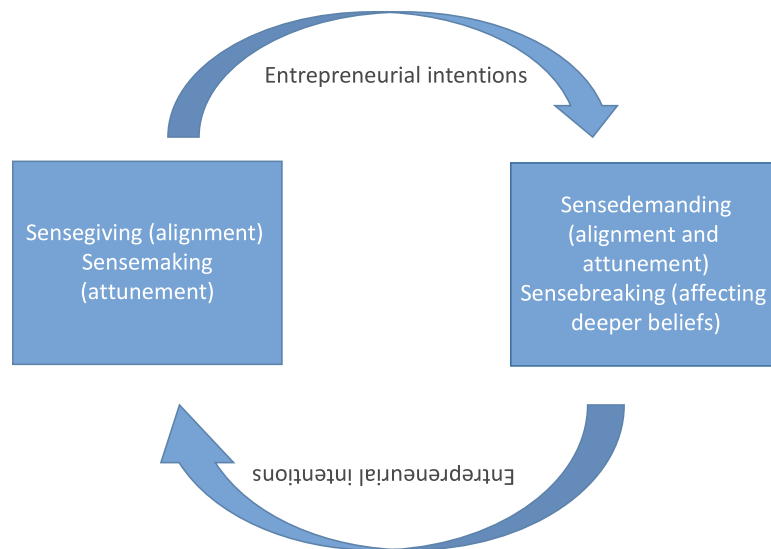


Fig. 10.2 Entrepreneurial intentions: a dynamic, bidirectional model using the sense-making perspective

... and Letting Feedback Shape Entrepreneurial Intentions

Entrepreneurs engage in feedback not only within the managerial team but also with other relevant stakeholders, for example investors and first customers (Clarke 2011; Cornelissen and Clarke 2010; Zott and Huy 2007; Santos and Eisenhardt 2005). In their pursuit for resources, all these parties are potential sources of sense-breaking. Three mechanisms of sense-breaking can be distinguished: redirecting, reframing and questioning.

Redirecting is associated with changes in one's business model and strategy as well as associated tactics (Kaffka 2017). Redirecting instances shape EIs regarding the 'what' and 'how' of opportunity development and exploitation (Kaffka 2017).

Questioning is the most direct form of negative critical feedback and is associated with direct criticism of an entrepreneur's existing assumptions regarding the entrepreneurial opportunity. Questioning occurrences can contribute to more awareness among entrepreneurs about what third parties see as erroneous or faulty in the way the entrepreneur legitimizes his or her venture idea.

Reframing complements questioning in the sense that it helps to not only critically examine one's own assumptions but is linked to challenging what Krueger (2007) describe as 'deep assumptions' the entrepreneur's taken-for-granted understanding and values regarding the opportunity they are developing. Reframing occurrences lead to an identity and attitude development among entrepreneurs because it encourages a change in the existing perspective, way of thinking and subsequent revaluations (Kaffka 2017).

Awareness of the different mechanisms of sense-breaking—and their effect on novel sense-making—provides entrepreneurs with a deeper command of the direction and purpose of feedback which is sought. We posit that entrepreneurship educators, but also trainers in incubation/acceleration programs, can make entrepreneurs more aware of the effects of sense-breaking as a tool to avoid the confirmation. For example, higher education institutions (HEIs) could host events at science parks and incubators, have entrepreneurs talk about their business idea and all the while engage in interaction that entails occurrences of sense-breaking activities, stimulate reflection on one's own values and assumptions and thereby create a more critical and open mind.

The link between EIs, the change in deeper beliefs and sense-breaking mechanisms is that certain deep beliefs or assumptions can be reframed which in turn affects the entrepreneur's intentions.

Entrepreneurial Intentions and Entrepreneurship Education

Rauch and Hulsink (2015) find that entrepreneurship education has a positive effect on attitudes, perceived behavioral control and the intention to become an entrepreneur. This shows that entrepreneurship education affects the personal attributes involved in following one's entrepreneurial intent. The authors also find that entrepreneurship education increasingly often involves context-oriented learning, such action-based or project-based learning (Rauch and Hulsink 2015) geared toward learning by doing.

Nabi et al. (2017) show that experiential learning—or learning by doing—affects EIs more than other types of learning. They undertook a systematic review of a range of entrepreneurship education impacts in higher education and observe that programs which have a more experiential pedagogical nature show impact at higher levels of EIs (actual start-ups and socio-economic impact), as opposed to more supply, supply-demand

and demand-oriented programs. The latter focus more upon basic or lower levels of the authors’ framework.

How can entrepreneurial education—which encompasses both entrepreneurship education and the development of entrepreneurial skills in the broadest way—contribute to stimulating a change in intentions into the direction of more willingness to or openness for entrepreneurial activity? Or more generally: what gets people to change or pivot their intentions? The question of how entrepreneurs react to negative feedback or barriers—and the impact of that reaction on their cognition or behavior—has been posed by Krueger (2009) who related perceived barriers to deeply held beliefs in which EIs are anchored. We know that the pedagogical tools that transformational educators use are based on changing deeper beliefs, or at least on raising awareness for and challenging one’s deeper beliefs. Practice seems to be clear on it: If we are going to change minds—which then are more ready and open for negative feedback—how are we are going to do that?

*Experiential Learning, Feedback Loops, Sense-Breaking:
And Entrepreneurial Intentions*

Since learning by doing invariably involves feedback loops, these can be intentionally used in order to avoid the confirmation bias. Indeed, we see that entrepreneurship education increasingly encourages this type of learning by drawing on experiential learning activities, for example, ideas from the lean start-up method (Nabi et al. 2017). That demonstrates the role of intentionality in feedback engagement—and associated sense-breaking occurrences—in entrepreneurship education.

Characteristically the widespread lean start-up method encourages entrepreneurs to seek feedback from relevant stakeholders to improve or adapt their proposed business model (Blank 2006). The sense-making perspective offers conceptual tools for the analysis and measure of everyday intentions which reflect the entrepreneurial intentionality.

In particular, sense-breaking leads the entrepreneur to change his perspective and formulate new intentions. Sense-breaking is like a cognitive push we feel, that even ‘hurts’ (as we hear our articulated thoughts not being confirmed but being redirected, reframed or even questioned). Sense-breaking does so by challenging existing assumptions (Pratt 2000; Vlaar et al. 2008), making entrepreneurs realize logical holes in their arguments, broadening their perspective and triggering new or different focus.

Subsequently the entrepreneurial idea can be improved, or new ideas can arise via critical feedback in the form of sense-breaking. In this way, sense-breaking facilitates the adding of new or different ‘dots’ when ‘connecting the dots’ for the realization of a viable business opportunity.

Considering those developments in entrepreneurship education, we suggest that the sense-making perspective offers valuable insights into the conceptualization of EIs and how to teach, learn and measure them in entrepreneurship education.

CONCLUSION

Science Is Hard

The trajectory of research—and application—that EI has taken us is far from over. Not only did it help bring serious social psychology research into the forefront of entrepreneurship, it continues to drive us in what is already valuable directions (and marvelous insights into entrepreneurial learning). In our efforts to understand ‘what lies beneath’ intentions, we see great value in ever-deepening theoretical and methodological opportunities. But these are not as simple as, say, TPB. Whether it is neuroscience or social ontology or even the sense-making perspective—with its related concept of sense-breaking—the bar is being raised. However, we have shared a most useful example that demonstrates how useful going deeper can be (and already is).

The sense-making perspective offers an angle for examining the role of EIs in entrepreneurship education. Sense-demanding and sense-breaking helps to conceptualize the dynamic nature of EIs. By engaging in feedback processes, stakeholders provide intersubjective knowledge to business opportunity development (Venkataraman et al. 2012) and are potential sources of sense-demanding, sense-breaking and sense-giving.

By introducing learning that is ‘rich’ in sense-breaking occurrences, entrepreneurship educators can help to counter biases in how aspiring entrepreneurs receive feedback and shape EIs. *This design of* entrepreneurship education maximizes interaction opportunities with stakeholders and subsequent feedback occurrences. And those intentional interactions in turn shape EIs.

As we argued, an intentional stance toward seeking (negative) feedback—and associated sense-breaking occurrences—helps to avoid the

confirmation bias. The entrepreneur's engagement with stakeholders to 'sell' their idea—how novice or maladroit that engagement may be—shapes an entrepreneurial mindset which benefits from intentionality regarding feedback that is sought and how this feedback is processed. Sense-making processes shape entrepreneurial/intentional actions such as seeking and receiving feedback, which in turn shape EIs.

We propose that by teaching entrepreneurship students how to seek and receive feedback intentionally, they can effectively combat the confirmation bias and discuss how the sense-making dimensions are tied to elements of attunement and alignment. The engagement in feedback loops inherent to problem-based learning explains why that type of learning impacts EIs; contributes to shaping EIs and perhaps, even the 'deeper' beliefs that underlie those intentions.

Recent educational models which emphasize 'learning by doing' fit clearly and illustrate the importance of feedback loops. In general, HEIs, and other educational facilities, have increasingly leveraged more experiential forms of learning such as problem-based learning (PBL).² Programs that stimulate experiential learning appear to have a higher chance of leading to the realization of new ventures (Nabi et al. 2016). In experiential or problem-based learning in entrepreneurship, interactions are specifically encouraged because they help to shape the opportunity, for example, interaction with and associated feedback from leading customers or in the beginning stage from friends and family.

Another aspect of intentional feedback is their merit in developing so-called twenty-first-century skills. The Organisation for Economic Co-operation and Development (OECD) defines twenty-first-century skills as skills for innovation, such as technical skills including disciplinary know-what and know-how; creativity skills such as curiosity, critical thinking, problem-solving and making connections; and social and behavioral skills such as interest, engagement, self-directed learning, self-confidence, organizational and communication competences, (cross-cultural) collaboration, teamwork and leadership.³ To develop most if not all of these skills necessitate engagement in feedback loops, for example with teachers and peers. Those skills are also reminiscent of skills which also play a role in entrepreneurship education. The scope of these skills suggests that more problem-based learning could be more appropriate in shaping an entrepreneurial mindset and associated twenty-first-century skills.

What Lies Beneath Entrepreneurial Intentions?

Possibly nothing less than the future of entrepreneurship research! A bold claim to be sure, but as we look more deeply into the underpinnings of entrepreneurial thinking and action, we are discovering powerful new ways to understand entrepreneurs and entrepreneurship. We are discovering new research tools. We are even headed toward becoming better educators.

And as we do, we are beginning to open doors outside of the entrepreneurship field. What can others learn from the rigorous studies of entrepreneurs? It is clear that much of the most innovative work on intentions in general is from our field, and the new methods like Fuzzy Set Qualitative Choice Analysis & Electro Encephalo Graph (fsQCA and EEG) play a key role. Science IS hard but the trajectory of research on EIs proves that the rewards are well worth it. Join us!

NOTES

1. We would also note that Per Davidsson attempted to develop a model of entrepreneurial intentions from first principles (1995); he found the same antecedents.
2. OECD. 2014. Promoting Skills for Innovation.
3. OECD. 2014. Promoting Skills for Education. Link: <http://repositorio.minedu.gob.pe/bitstream/handle/123456789/2482/Promoting%20Skills%20for%20Innovation%20in%20Higher%20Education%20A%20Literature%20Review%20on%20the%20Effectiveness%20of%20Problem-Based%20Learning%20and%20of%20Teaching%20Behaviours.pdf?sequence=1>

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